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ORIGINAL ARTICLE

Association Between Postprandial Hypotension Determined by Ambulatory Blood Pressure Monitoring and Falls Among Older Adults With Hypertension Who Are Taking Antihypertensive Medication: Results From the AMBROSIA Study

Keisuke Narita, C. Barrett Bowling, Kimberly Cannavale, Chloe Fang, Teresa N. Harrison, Paul Muntner, Lei Qian, Joseph E. Schwartz, John J. Sim, Rong Wei, Kristi Reynolds, Daichi Shimbo

BACKGROUND: Postprandial hypotension (PPH) may contribute to falls among older adults, particularly those taking antihypertensive medication. However, evidence on this association in community-dwelling populations is limited. Since ambulatory blood pressure (BP) monitoring captures BP during daily activities, it may provide accurate assessments of PPH outside the clinic setting.

METHODS: This prospective cohort study examined the association between PPH and fall risk among community-dwelling adults aged ≥65 years taking antihypertensive medication. At baseline, participants underwent 24-hour ambulatory BP monitoring; subsequently, they completed monthly fall calendars during a 12-month follow-up. PPH by systolic BP (SBP; systolic PPH) was defined as a postprandial SBP decline, mean SBP during the hour before the meal minus the minimum SBP during the 2 hours after the meal, following any meal of ≥20 mmHg, or a decrease to SBP ≤90 mmHg when preprandial SBP was ≥100 mmHg.

RESULTS: Among 626 participants (mean±SD age, 74.6±6.2 years; 56.1% women), 442 (70.6%) experienced systolic PPH. The mean±SD number of meals was 2.6±0.8 during the ambulatory BP monitoring period. During the 12-month follow-up, falls occurred in 169 of 442 (38.2%) participants with systolic PPH and 70 of 184 (38.0%) participants without systolic PPH. Systolic PPH was not associated with fall risk (adjusted hazard ratio, 0.93 [95% CI, 0.69–1.26]). A restricted cubic spline analysis demonstrated no evidence of an association between the largest postprandial SBP decline across all meals and fall risk.

CONCLUSIONS: In this cohort study, PPH identified by ambulatory BP monitoring was common but not associated with risk of falls. ***(Hypertension.* 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.25518.)** • **Supplement Material**.

Key Words: aged ■ blood pressure ■ humans ■ hypotension ■ meals

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Falls among older adults are associated with an increased risk of disability and mortality.[1–3] A fall has been reported to increase the 1-year mortality risk 3-fold among individuals aged ≥65 years.[1] Nearly two-thirds of US adults aged ≥65 years have hypertension,[4] and 32% of older adults with hypertension experience falls annually.[5] Use of antihypertensive medications has been associated with an increased risk of fall injuries.[6] Identifying risk factors for falls in older adults taking antihypertensive medication may facilitate the development of interventions to reduce this risk.[7]

Postprandial hypotension (PPH), defined as a large reduction in blood pressure (BP) following a meal, is common in older adults.[8–10] In prior studies, 55% to 73%

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NOVELTY AND RELEVANCE

What Is New?

Postprandial hypotension determined by ambulatory blood pressure monitoring was common among community-dwelling older adults who were taking antihypertensive medication.

In this cohort, postprandial hypotension determined by ambulatory blood pressure monitoring was not associated with falls risk.

What Is Relevant?

Hypotension is a recognized adverse effect among older patients with hypertension taking antihypertensive medications and has been reported to increase fall risk. Identifying if postprandial hypotension is associated with fall risk can provide direction for interventions to reduce fall risk among older adults with hypertension.

Clinical/Pathophysiological Implications?

Postprandial hypotension determined by ambulatory blood pressure monitoring is unlikely to serve as a practical marker for identifying older patients taking antihypertensive medication who are at high risk of falls.

Nonstandard Abbreviations and Acronyms

ABPM	ambulatory blood pressure monitoring
AMBROSIA	Ambulatory Blood Pressure in Older Adults
BP	blood pressure
DBP	diastolic blood pressure
HR	hazard ratio
PPH	postprandial hypotension
SBP	systolic blood pressure

of adults with hypertension experienced PPH.[10–12] Since a sudden BP decrease may lead to dizziness or syncope, PPH is often thought to predispose older adults to falls.[8,9] Studies among nursing home residents and among patients followed after hospital discharge have reported an association between PPH and increased fall risk.[13,14] However, these studies included select populations and used standardized meals for determining PPH.[13,14] PPH and postprandial BP decline in community-dwelling populations may be better evaluated by measuring BP after regular meals outside the clinic in an individual's naturalistic setting.

Ambulatory BP monitoring (ABPM) provides 24-hour BP profiles during daily activities and is recommended as an adjunct to office BP for hypertension diagnosis and management.[15,16] Despite this recommendation, data on the association between PPH determined by ABPM and fall risk among community-dwelling older adults are scarce. We hypothesized that PPH assessed by ABPM is associated with increased fall risk among older adults taking antihypertensive medication. Confirmation of this hypothesis would indicate that ABPM may be useful for evaluating PPH to identify treated older adults at increased risk of falls. In the current study, we investigated the association between PPH determined by ABPM and the risk of falls among community-dwelling adults aged ≥65 years taking antihypertensive medication.

METHODS

Data are available upon reasonable request. Anonymized data may be made available from K. Reynolds and D. Shimbo upon reasonable request from qualified researchers with documented evidence of training for human subjects protections and upon approval of analysis proposals with signed data-access agreements in place

Study Design

The AMBROSIA study (Ambulatory Blood Pressure in Older Adults) was designed to determine whether ABPM can identify individuals who are at increased risk for falls among older adults with hypertension taking antihypertensive medication. Details of the protocol for this prospective cohort study have been published.[17] In brief, participants were enrolled between May 2019 and November 2022 at Kaiser Permanente Southern California, which provides comprehensive health care services to >4.8 million residents of Southern California.[18] Eligible participants were identified from >123 000 Kaiser Permanente Southern California members aged ≥65 years who were being treated for hypertension. Individuals who were nonambulatory, had a history of dementia, or had resided in a health care facility within the 12 months preceding recruitment were not eligible for enrollment. Detailed inclusion and exclusion criteria are in Table S1. Study staff screened 1068 individuals by telephone and scheduled 948 for an in-person visit (Visit 1) at the research clinic. Of these individuals, 670 were eligible, remained interested, and were enrolled. Among the enrolled participants, 656 completed 24-hour ABPM. For the current analysis, participants who had insufficient BP readings to determine PPH (n=4), did not report having a meal during the ABPM period (n=2), or had missing falls calendars (n=24) were excluded. The final analytic sample comprised

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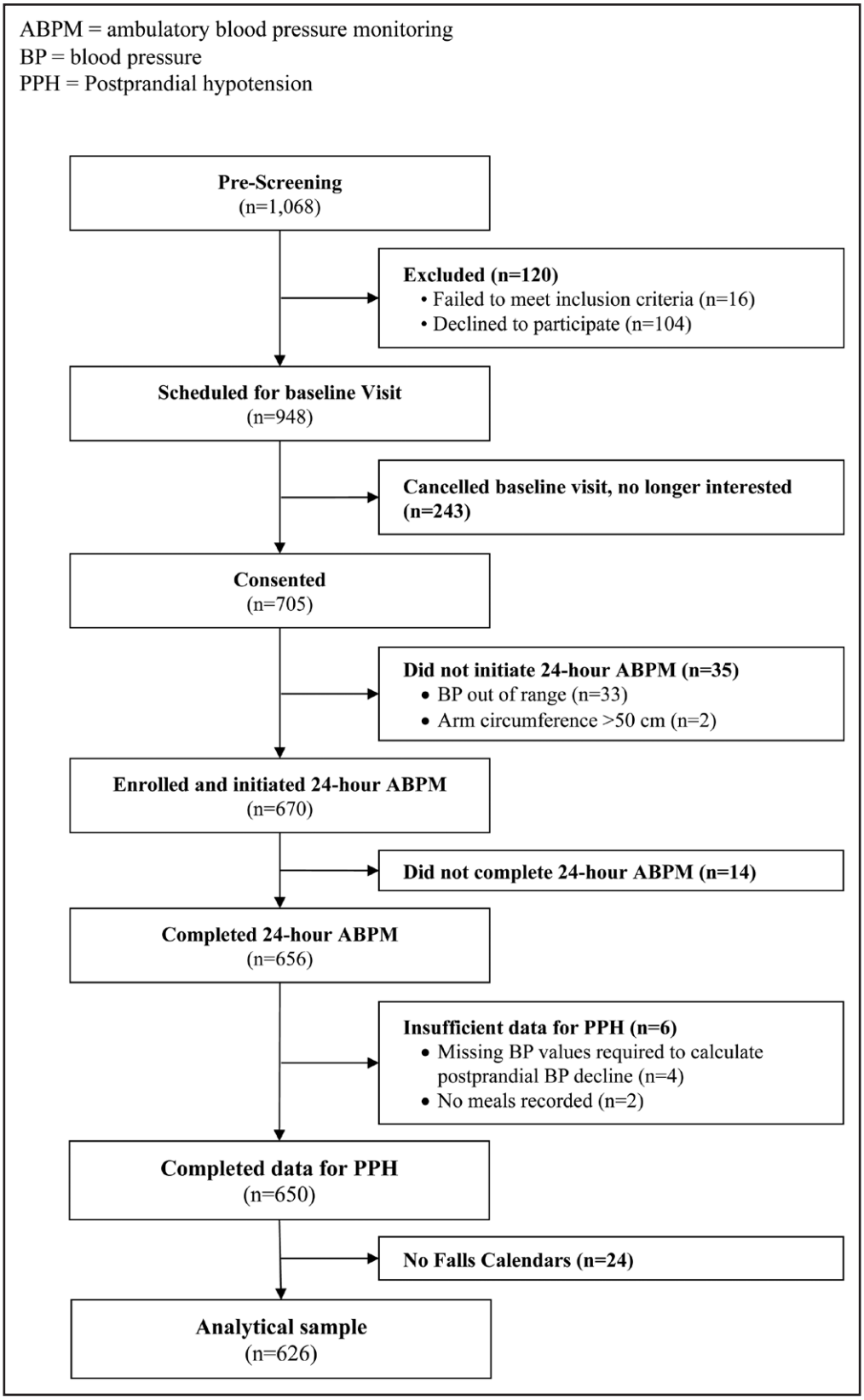

Figure 1. Assembly of analytical sample.

626 participants (Figure 1). The study was approved by the KPSC and Columbia University Irving Medical Center institutional review boards.

Study Procedures and BP Measurements

Demographics, clinical characteristics, and geriatric factors (ie, frailty, cognition, and physical performance) were

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assessed, and research-grade clinic BP and 24-hour ABPM measurements were performed at Visit 1 and a subsequent visit the next day (Visit 2). Information obtained from the electronic health record included laboratory results and medical history (eg, diabetes and cardiovascular disease, based on diagnosis codes). Prediabetes was defined as the absence of diabetes and the presence of at least one of the following: a diagnosis code of prediabetes; HbA1c of 5.7% to <6.5%; or fasting plasma glucose of 100 mg/dL to <126 mg/dL. Chronic kidney disease was defined as an estimated glomerular filtration rate <60 mL/min per 1.73 m^2. The details of the study procedures are provided in Table S2. Clinic BP values obtained as part of an outpatient visit within 12 weeks before enrollment were extracted from the electronic health record. These measurements were performed by trained medical staff using a validated oscillometric device, Accutorr 7 (Mindray DS USA, Inc).[19,20]

Research-grade clinic BP was measured in a seated position using a validated oscillometric device, HEM-907XL (Omron Healthcare, Japan).[21,22] The mean research-grade clinic BP was the average of 3 BP readings taken at 1-minute intervals on the nondominant arm. Orthostatic hypotension was also assessed at Visit 1.[23] For 24-hour ABPM, participants were fitted with an appropriately sized cuff on the nondominant arm using a validated device, SpaceLabs 90227 (Spacelabs Healthcare, Snoqualmie, WA).[24] Participants were instructed to record the times of sleep onset and awakening, and the start and end times of each meal and all naps, using the paper-based log. BP readings on ABPM were taken every 20 minutes during the daytime and every 30 minutes during the nighttime.[25,26] ABPM data were used for the current analysis if there was at least 1 BP reading during the hour before and at least 1 BP reading within 2 hours after at least one meal. Participants and their health care providers were not provided the research-grade clinic and ABPM BP values to avoid influencing treatment decisions. Additional details are provided in Supplemental Material.

Definition of PPH

Ambulatory BP values and mealtimes from the ABPM log were used to assess PPH. For each reported meal, a postprandial systolic BP (SBP) decline was calculated as the difference between mean SBP during the hour before the meal, referred to as the preprandial SBP, and the lowest SBP recorded during the 2 hours after the end of the meal, referred to as the postprandial SBP. In the primary analyses, systolic PPH was defined as a preprandial minus postprandial SBP difference ≥20 mm Hg or a postprandial SBP ≤90 mm Hg when the mean preprandial SBP was ≥100 mm Hg.[8,9,11] In secondary analyses, diastolic PPH was defined as a preprandial minus postprandial diastolic BP (DBP) ≥13 mm Hg following any meal. As a validated definition for diastolic PPH has not been reported, the threshold was determined from the relationship between SBP and DBP observed in a prior study (Supplemental Material).[27] Asleep BP readings that occurred during either the preprandial or postprandial period were excluded from the calculations of PPH. Based on several prior studies, an alternative definition, in which PPH was defined using the lowest SBP recorded during the 1-hour period after the end of the meal, was used in a sensitivity analysis.[28,29]

Assessment of Falls

Data on falls over the 12 months following Visit 2 were collected prospectively using monthly falls calendars. This method has been validated and used successfully in prospective long-term (>12 months) studies of falls.[30–32] Participants were provided with a set of monthly calendars and instructions to record falls when they occurred and to return the falls calendars at the end of each month. Participants were asked to record any fall where part of your body hits a lower surface, including falls that occur on stairs, by marking each day that a fall occurred on the calendar. Participants were asked to provide information on the timing and location of the fall, and whether they required medical care because of the fall. The primary outcome was time from Visit 2 to the first fall. Additional details on the ascertainment of falls are provided in Supplemental Material.

Statistical Analysis

Summary statistics for demographic and clinical characteristics were estimated overall, as well as for those with and without systolic PPH. The Kruskal-Wallis test and Fisher exact test were used to compare continuous and categorical measures, respectively, of participant characteristics between those with and without systolic PPH. Fall incidence was calculated per 100 person-years for those with and without systolic PPH. Cox proportional hazards models estimated hazard ratios (HRs) and 95% CIs for time to first fall, comparing participants with and without systolic PPH. The adjusted Cox models included the following covariates: Model 1 included age, sex, and race/ethnicity; Model 2 added body mass index, diabetes, number of antihypertensive medication classes, alcohol use, impaired physical performance based on the short physical performance battery score, number of other supplied chronic medication classes, history of falls, chronic kidney disease, and orthostatic hypotension; and Model 3 further included the number of meals for which PPH could be evaluated. Stratified analyses were done by age (65–74 years or ≥75 years); sex; race and ethnicity; self-reported history of falls in the year before enrollment (yes or no); high clinic BP from the electronic health record defined as SBP ≥140 mm Hg and DBP ≥90 mm Hg (yes or no); and short physical performance battery assessment score (score 0–9 or 10–12). These analyses were repeated for diastolic PPH. Adjusted Cox models with restricted cubic splines were also used to assess associations with falls for the largest postprandial SBP and DBP changes across all available meals as a continuous variable. In a sensitivity analysis, PPH, calculated from BP readings excluding those obtained during naps, was used to calculate incidence rates and HRs for falls with the aforementioned models. The timing of naps during 24-hour ABPM was based on self-report. In addition, in a sensitivity analysis, we calculated incidence rates and HRs for falls using the aforementioned models, defining PPH by the lowest SBP within the 1-hour period after a meal. All analyses were performed using SAS, version 9.4 (SAS Institute, Inc, Cary, NC) and R software, version 4.1.3 (R Foundation for Statistical Computing, Vienna, Austria). A 2-sided $P<0.05$ was considered statistically significant.

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RESULTS

Systolic PPH

Participant Characteristics, and the Association of Systolic PPH With the Risk of a Fall

Among the 626 participants, the mean (SD) age was 74.6 (6.2) years; 56.1% were women, and 44.4% were non-Hispanic White (Table 1). The mean (SD) number of meals during the 24-hour ABPM period was 2.6 (0.8). Overall, 442 (70.6%) participants had systolic PPH. Compared with those without systolic PPH, participants with systolic PPH were older (75.1 years versus 73.4 years) and had higher mean research-grade clinic SBP (132.3 mm Hg versus 129.3 mm Hg).

During the 1-year follow-up, 239 (38.2%) participants reported experiencing a fall: 70 (38.0%) of the 184 participants without systolic PPH and 169 (38.2%) of the 442 participants with systolic PPH. The incidence rates (95% CI) were 53.3 (42.1–67.3) and 53.4 (46.0–62.1) per 100 person-years in the participants without and with systolic PPH, respectively. The HR (95% CI) for the risk of a fall associated with systolic PPH was 1.00 (0.76–1.33) in the unadjusted model; 0.91 (0.69–1.21) in Model 1; 0.93 (0.70–1.25) in Model 2; and 0.93 (0.69–1.26) in Model 3. Participants with systolic PPH did not have an increased risk of falls when investigated in subgroups defined by age; sex; race and ethnicity; self-reported history of falls; high clinic BP; or short physical performance battery assessment score (Table 2).

Diastolic PPH

Participant Characteristics, and the Association of Diastolic PPH With the Risk of a Fall

Of the 626 participants, 465 (74.3%) had diastolic PPH. Participants with diastolic PPH, compared with those without diastolic PPH, had higher mean research-grade clinic DBP (70.0 mm Hg versus 66.3 mm Hg), awake DBP on ABPM (74.1 mm Hg versus 71.8 mm Hg), and 24-hour DBP on ABPM (71.1 mm Hg versus 69.2 mm Hg). There was no evidence of other differences in baseline characteristics between participants with and without diastolic PPH (Table S3). Compared with participants without diastolic PPH, those with diastolic PPH did not have an increased risk for a fall in the unadjusted model or any of the adjusted models, either in the overall sample or within any of the analyzed subgroups (Table S4).

Spline Analyses for the Associations of the Largest Postprandial SBP and DBP Decline With the Risk of a Fall

The restricted cubic spline analyses for the associations of the largest postprandial SBP and DBP with the risk of a fall showed that the 95% CI always included the reference value (HR=1) for both spline curves (Figure 2A and 2B). The likelihood ratio test showed no linear or curvilinear (spline model) association of either the largest postprandial SBP or largest postprandial DBP with the risk of a fall.

Sensitivity Analyses

In the sensitivity analysis using systolic PPH calculated from BP readings excluding those obtained during naps, 433 (69.2%) participants had systolic PPH. The HR (95% CI) for the risk of a fall associated with systolic PPH was 0.97 (0.72–1.31) in Model 3 (Table S5). In the sensitivity analysis using the alternative definition of PPH, defined as the lowest SBP during the 1-hour period after the meal, 353 (56.7%) participants had systolic PPH, and the HR (95% CI) for the risk of a fall associated with systolic PPH was 1.03 (0.78–1.36) in Model 3 (Table S6).

DISCUSSION

Among community-dwelling adults aged ≥65 years with hypertension who were taking antihypertensive medication, more than two-thirds had PPH when assessed by ABPM. However, neither systolic nor diastolic PPH assessed by ABPM was associated with an increased risk of a fall. Further, neither the largest postprandial SBP decline nor the largest postprandial DBP decline was associated with an increased risk of a fall. These findings suggest that PPH is common among community-dwelling older adults taking antihypertensive medication, but that ABPM-identified PPH is not associated with risk of falls.

The 2025 American College of Cardiology/American Heart Association BP guideline states that using ABPM may improve detection of symptomatic hypotension.[15] The European Society of Hypertension position paper also recommends ABPM for evaluating excessive BP decreases, including postural hypotension and PPH, among older adults.[33] However, most prior studies used BP measured in medical facilities and a standardized diet to evaluate PPH.[10,34,35] In the systematic review and meta-analysis by Huang et al, [10] which examined the prevalence of PPH among older adults, none of the 13 identified studies used ABPM for out-of-office BP measurement to detect PPH. Jenkins et al[35] conducted a systematic review and meta-analysis to investigate the association between PPH and cardiovascular prognosis. Of the 7 studies analyzed, only 1 used ABPM for out-of-office BP measurement to detect PPH.[11]

A few studies have examined the association between PPH and fall risk. A systematic review by Jansen et al identified 4 observational studies, including 3 cross-sectional and 1 prospective[13,36–38] that examined the

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Table 1. Baseline Characteristics Overall and Among Participants With and Without Systolic Postprandial Hypotension

	Total (N=626)	Participants without systolic PPH (N=184)	Participants with systolic PPH (N=442)	*P* value
Mean (SD) age at visit 1, yr	74.6 (6.2)	73.4 (5.8)	75.1 (6.3)	0.002*
Age categories at visit 1, n (%)				0.035†
65–70 yr	180 (28.8%)	65 (35.3%)	115 (26.0%)	
71–75 yr	193 (30.8%)	62 (33.7%)	131 (29.6%)	
76–80 yr	149 (23.8%)	35 (19.0%)	114 (25.8%)	
81–85 yr	66 (10.5%)	14 (7.6%)	52 (11.8%)	
≥85 yr	38 (6.1%)	8 (4.3%)	30 (6.8%)	
Female sex, n (%)	351 (56.1%)	100 (54.3%)	251 (56.8%)	0.597†
Race and ethnicity, n (%)				0.115†
Non-Hispanic White	278 (44.4%)	67 (36.4%)	211 (47.7%)	
Non-Hispanic Asian or Pacific Islander	83 (13.3%)	27 (14.7%)	56 (12.7%)	
Non-Hispanic Black	140 (22.4%)	49 (26.6%)	91 (20.6%)	
Hispanic	114 (18.2%)	37 (20.1%)	77 (17.4%)	
Other or unknown	11 (1.8%)	4 (2.2%)	7 (1.6%)	
Body mass index, n (%)				0.252†
Underweight or normal weight: <24.9 kg/m^2	155 (24.8%)	35 (19.0%)	120 (27.1%)	
Overweight: 25–29.9 kg/m^2	242 (38.7%)	77 (41.8%)	165 (37.3%)	
Obese: 30–34.9 kg/m^2	150 (24.0%)	46 (25.0%)	104 (23.5%)	
Severely obese: 35–39.9 kg/m^2	58 (9.3%)	18 (9.8%)	40 (9.0%)	
Morbidly obese: ≥40 kg/m^2	21 (3.4%)	8 (4.3%)	13 (2.9%)	
Smoking, n (%)				0.684†
Current	11 (1.8%)	4 (2.2%)	7 (1.6%)	
Former	232 (37.3%)	65 (35.3%)	167 (38.1%)	
Never	379 (60.9%)	115 (62.5%)	264 (60.3%)	
Alcohol use, n (%)	387 (62.1%)	109 (59.6%)	278 (63.2%)	0.415†
Diabetes, n (%)	190 (30.4%)	62 (33.7%)	128 (29.0%)	0.253†
Prediabetes, n (%)	253 (40.4%)	69 (37.5%)	184 (41.6%)	0.372†
Mean (SD) HbA1c, %	6.2 (1.1)	6.4 (1.3)	6.1 (1.0)	0.135*
History of coronary heart disease, n (%)	108 (17.3%)	27 (14.7%)	81 (18.3%)	0.297†
History of heart failure, n (%)	25 (4.0%)	6 (3.3%)	19 (4.3%)	0.658†
Chronic kidney disease, n (%)	144 (23.2%)	44 (24.2%)	100 (22.8%)	0.754†
Mean (SD) eGFR, ml/min per 1.73 m^2	72.2 (18.0)	71.3 (18.2)	72.6 (17.9)	0.353*
History of neuropathy, n (%)	70 (11.2%)	20 (10.9%)	50 (11.3%)	1.000†
History of depression, n (%)	115 (18.4%)	35 (19.0%)	80 (18.1%)	0.821†
History of Parkinson's disease, n (%)	6 (1.0%)	2 (1.1%)	4 (0.9%)	1.000†
History of arthritis, n (%)	286 (45.7%)	84 (45.7%)	202 (45.7%)	1.000†
Charlson comorbidity index score, n (%)				0.446†
0	104 (16.6%)	37 (20.1%)	67 (15.2%)	
1	140 (22.4%)	37 (20.1%)	103 (23.3%)	
2	120 (19.2%)	33 (17.9%)	87 (19.7%)	
≥3	262 (41.9%)	77 (41.8%)	185 (41.9%)	
No. of antihypertensive medication classes, n (%)				0.113†
0	6 (1.0%)	0 (0.0%)	6 (1.4%)	
1	218 (34.8%)	67 (36.4%)	151 (34.2%)	
2	254 (40.6%)	82 (44.6%)	172 (38.9%)	

(*Continued*)

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Table 1. Continued

	Total (N=626)	Participants without systolic PPH (N=184)	Participants with systolic PPH (N=442)	*P* value
≥3	148 (23.6%)	35 (19.0%)	113 (25.6%)	
Number of other medication classes, n (%)				0.864†
0–1	217 (34.7%)	62 (33.7%)	155 (35.1%)	
2–3	265 (42.3%)	81 (44.0%)	184 (41.6%)	
≥4	144 (23.0%)	41 (22.3%)	103 (23.3%)	
Use of assistive devices, n (%)	109 (17.5%)	28 (15.4%)	81 (18.4%)	0.215†
Difficulty with >1 ADL or IADL, n (%)	76 (12.2%)	19 (10.3%)	57 (12.9%)	0.421†
MoCA Score, n (%)				0.557†
<26: mild cognitive impairment	443 (71.9%)	134 (73.6%)	309 (71.2%)	
≥26: no cognitive impairment	173 (28.1%)	48 (26.4%)	125 (28.8%)	
Mean (SD) life-space mobility score	82.3 (22.8)	81.0 (23.5)	82.9 (22.6)	0.319*
Mean (SD) total short physical performance battery assessment score	9.5 (2.0)	9.7 (1.8)	9.4 (2.0)	0.108*
Frailty, n (%)				0.702†
Robust: frailty score 0	286 (45.7%)	87 (47.3%)	199 (45.0%)	
Pre-frailty: frailty score 1 or 2	313 (50.0%)	91 (49.5%)	222 (50.2%)	
Frail: frailty score ≥3	27 (4.3%)	6 (3.3%)	21 (4.8%)	
Self-reported history of fall, n (%)	246 (39.9%)	75 (41.2%)	171 (39.3%)	0.719†
Mean (SD) SBP, mm Hg				
Clinic SBP	132.8 (12.5)	131.7 (11.8)	133.2 (12.8)	0.282*
Research grade clinic SBP	131.4 (14.7)	129.3 (14.3)	132.3 (14.8)	0.009*
Awake SBP on ABPM	134.6 (13.7)	133.1 (13.9)	135.2 (13.6)	0.087*
Asleep SBP on ABPM	121.1 (15.6)	120.7 (16.1)	121.3 (15.3)	0.439*
24-hour SBP on ABPM	130.1 (13.1)	128.9 (13.8)	130.6 (12.8)	0.118*
Mean (SD) DBP, mm Hg				
Clinic DBP	70.0 (9.6)	69.9 (9.9)	70.1 (9.5)	0.923*
Research grade clinic DBP	69.1 (10.3)	68.5 (9.9)	69.3 (10.4)	0.388*
Awake DBP on ABPM	73.5 (8.5)	72.7 (8.0)	73.8 (8.7)	0.181*
Asleep DBP on ABPM	64.4 (8.6)	63.8 (8.4)	64.7 (8.7)	0.380*
24-hour DBP on ABPM	70.6 (7.8)	69.7 (7.4)	71.0 (8.0)	0.153*
Orthostatic hypotension, n (%)	105 (16.9%)	23 (12.6%)	82 (18.7%)	0.078†

ABPM indicates ambulatory blood pressure monitoring; ADL, activities of daily living; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; IADL, instrumental ADL; MoCA, Montreal Cognitive Assessment; PPH, postprandial hypotension; and SBP, systolic blood pressure.

*Kruskal-Wallis test *P* value.

†Fisher exact test *P* value. Systolic PPH was defined as postprandial SBP decline ≥20 mm Hg or a decrease to ≤90 mm Hg within 2 h postprandially when preprandial SBP was ≥100 mm Hg at any meal (for which postprandial SBP decline was assessed). Chronic kidney disease is defined as eGFR <60 mL/min per 1.73 m^2.

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association of PPH with fall risk.[34] The 3 cross-sectional studies reported a higher prevalence of PPH among participants with versus without a self-reported or documented history of falls.[36–38] In a prospective study by Aronow et al,[13] the prevalence of systolic PPH at lunch was 24% of 499 nursing home residents. Postprandial SBP decline at lunch was associated with greater fall risk during a mean follow-up of 29 months (adjusted HR, 1.19 [95% CI, 1.15–1.23]).[39] A small study by Ali Abdelhamid et al not included in the Jansen et al[14] systematic review found systolic PPH, evaluated by clinic BP measurements before and after taking 75 g glucose, in 28% of 22 older adults who had been discharged from a general intensive care unit, and those with systolic PPH had an elevated risk of falls during the subsequent 12 months (unadjusted relative risk, 3.7 [95% CI, 1.6–8.8]). In contrast, the current study revealed a high prevalence of PPH yet found no association with fall risk. This discrepancy may be due to differences in BP measurement methodology and study populations. While the aforementioned prior studies relied on office BP monitoring, the current study assessed PPH after nonstandard meals

Table 2. Association of Systolic Postprandial Hypotension With the Risk of a Fall, Overall and Within Subgroups

	Participants without systolic PPH				Participants with systolic PPH				Hazard ratio (95% CI)			
	N	Number of falls	Number of person-years	Incidence per 100 person-years (95% CI)	N	Number of falls	Number of person-years	Incidence per 100 person-years (95% CI)	Unadjusted	Model 1	Model 2	Model 3
Overall	184	70	131.4	53.3 (42.1–67.3)	442	169	316.3	53.4 (46.0–62.1)	1.00 (0.76–1.33)	0.91 (0.69–1.21)	0.93 (0.70–1.25)	0.93 (0.69–1.26)
Subgroups												
Age at index date, yr												
65–74 yr	116	44	83.3	52.8 (39.3–70.9)	220	72	166.2	43.3 (34.4–54.6)	0.83 (0.57–1.20)	0.78 (0.54–1.14)	0.75 (0.50–1.13)	0.78 (0.51–1.17)
≥75 yr	68	26	48.1	54.1 (36.8–79.4)	222	97	150.1	64.6 (53.0–78.8)	1.19 (0.77–1.83)	1.09 (0.71–1.69)	1.19 (0.76–1.87)	1.14 (0.71–1.82)
Sex												
Female	100	41	69.7	58.8 (43.3–79.9)	251	103	178.4	57.7 (47.6–70.0)	0.98 (0.68–1.41)	0.87 (0.60–1.26)	0.87 (0.59–1.27)	0.88 (0.59–1.29)
Male	84	29	61.7	47.0 (32.6–67.6)	191	66	137.9	47.9 (37.6–60.9)	1.02 (0.66–1.58)	0.95 (0.61–1.48)	1.02 (0.62–1.67)	1.02 (0.62–1.68)
Race and ethnicity												
Non-Hispanic White	67	32	47.7	67.1 (47.5–94.9)	211	101	138.9	72.7 (59.8–88.4)	1.06 (0.71–1.58)	1.01 (0.68–1.51)	1.12 (0.73–1.72)	1.06 (0.69–1.64)
Non-Hispanic Asian/Pacific Islander	27	8	20.7	38.6 (19.3–77.2)	56	14	47.6	29.4 (17.4–49.6)	0.77 (0.32–1.84)	0.77 (0.32–1.85)	0.31 (0.09–1.10)	0.31 (0.07–1.37)
Non-Hispanic Black	49	15	36.3	41.4 (24.9–68.6)	91	28	67.2	41.7 (28.8–60.3)	1.01 (0.54–1.88)	1.00 (0.53–1.89)	0.84 (0.41–1.74)	0.83 (0.39–1.77)
Hispanic	37	14	23.5	59.5 (35.2–100.5)	77	25	55.4	45.1 (30.5–66.7)	0.76 (0.40–1.46)	0.67 (0.34–1.33)	0.66 (0.31–1.43)	0.74 (0.34–1.62)
Self-reported fall												
Yes	75	40	44.9	89.1 (65.3–121.4)	171	85	105.1	80.8 (65.4–100.0)	0.92 (0.63–1.33)	0.91 (0.62–1.33)	0.88 (0.60–1.31)	0.91 (0.61–1.35)
No	107	28	85.4	32.8 (22.7–47.5)	264	81	206.4	39.2 (31.6–48.8)	1.19 (0.78–1.83)	1.04 (0.67–1.62)	1.04 (0.65–1.64)	1.04 (0.64–1.69)
High clinic SBP/DBP (≥140/90 mm Hg)												
Yes	41	13	30.5	42.6 (24.7–73.4)	108	41	77.0	53.2 (39.2–72.3)	1.22 (0.65–2.27)	1.00 (0.52–1.91)	0.88 (0.43–1.81)	1.02 (0.47–2.22)
No	143	57	100.9	56.5 (43.6–73.2)	334	128	239.3	53.5 (45.0–63.6)	0.95 (0.70–1.30)	0.95 (0.70–1.30)	0.92 (0.66–1.28)	0.91 (0.65–1.27)
Short physical performance battery assessment score												
0–9	76	31	53.0	58.6 (41.2–83.3)	212	80	149.1	53.7 (43.1–66.8)	0.91 (0.60–1.37)	0.83 (0.55–1.27)	0.90 (0.58–1.40)	0.91 (0.58–1.43)
10–12	108	39	78.5	49.7 (36.3–68.0)	230	89	167.2	53.2 (43.3–65.5)	1.07 (0.74–1.56)	0.98 (0.67–1.43)	0.99 (0.66–1.48)	0.96 (0.63–1.46)

The subgroup analysis in the race of other or unknown was not assessed because the small number of participants (n=11). In the subgroup analysis of self-reported fall, 9 participants were excluded by missing data for self-reported fall. Model 1: age, sex, race, and ethnicity. Model 2: Model 1 variables + body mass index, diabetes, alcohol use (excluding 3 missing cases), number of antihypertensive medication classes, number of other chronic medication classes, chronic kidney disease (eGFR <60 mL/min per 1.73 m^2; excluding 6 missing cases), short physical performance battery score, history of falls (excluding 9 missing cases), and orthostatic hypotension (excluding 4 missing cases). Model 3: Model 2 variables + number of meals. In Model 3, total 20 participants were excluded due to missing data. DBP indicates diastolic blood pressure; eGFR, estimated glomerular filtration rate; PPH, postprandial hypotension; and SBP, systolic blood pressure.

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eaten in one's naturalistic setting using ABPM. These measurements may enhance the ecological validity of the results. They also reflect the planning and adaptive behavior of participants, for whom the timing of meals and postprandial activity were self-determined and possibly influenced by their own biofeedback. In addition,

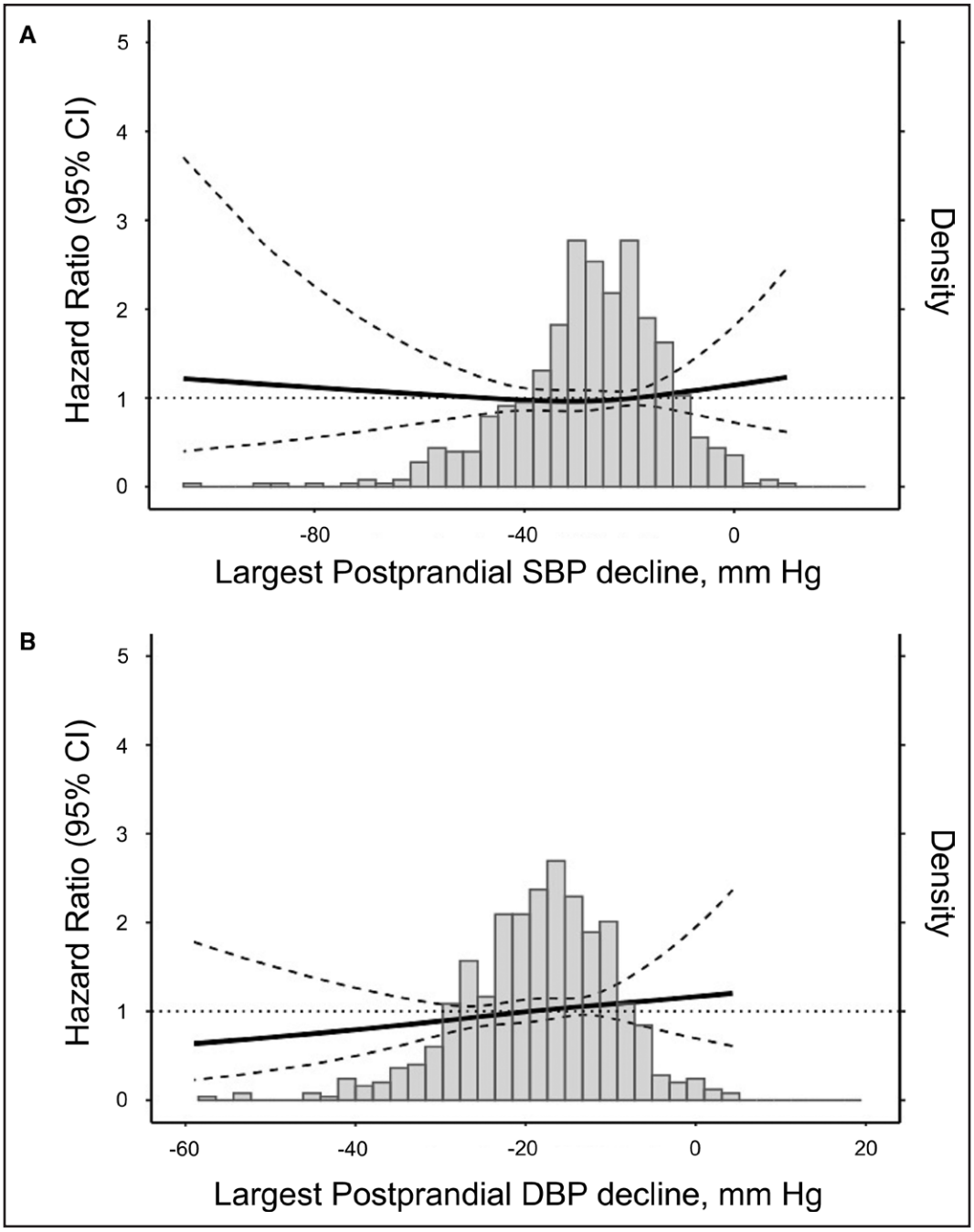

Figure 2. Hazard ratio for falls associated with the largest postprandial systolic and diastolic decline (n=626).
A, Spline curve depicting the hazard ratio between the largest postprandial systolic blood pressure (SBP) decline and the risk of first falls. **B**, Spline curve depicting the hazard ratio between the largest postprandial diastolic blood pressure (DBP) decline and the risk of first falls. The mean value in the analytic sample was used as reference and the number of knots was 3, selected on the basis of Akaike information criterion. These analytic models included confounding factors as the same factors in the multivariable Cox model, model 3 in Table 2. Solid line represents the hazard ratio, and dashed lines indicate the 95% CI. The dotted line is a hazard ratio of 1. The histogram shows the distribution of participants across the range of the largest postprandial SBP and DBP declines.

prior studies included frail, institutionalized, or recently hospitalized adults, whereas the current study participants were living independently and were comparatively less frail. Prediabetes and diabetes have been reported to be associated with elevated risks of frailty and falls.[40,41] Furthermore, prior studies have reported an associations of prediabetes and diabetes with PPH via autonomic dysfunction.[8,9,42,43] However, in the current analysis, the prevalence of prediabetes and diabetes did not differ between participants with and without PPH.

There are several strengths of the current study. We prospectively examined the association between PPH,

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assessed in a naturalistic setting outside the office using ABPM, and the fall risk. With an enrollment sample size of 670, the study was designed a priori to have adequate power (≥80%) to detect a minimum HR of 1.45 (2-tailed, α=0.05 test), assuming a 40% fall rate in those without PPH.[17] Furthermore, ABPM was conducted using a standardized protocol, and participants consumed their regular meals while undergoing ABPM. The treating clinicians were blinded to the research grade BP and ABPM results. Therefore, these results did not inform clinical decision-making regarding antihypertensive medication titration. The study also incorporated reference-standard assessments of falls, using prospectively assessed falls calendars that allowed examination of all falls, whether injurious or not, and geriatric fall-related factors.[30–32] There are also several potential limitations. In the current analysis, only 4.3% of participants had frailty, which is lower than estimates for older adults with hypertension.[44] Furthermore, given that all participants were covered by medical insurance, the study may not have included socially vulnerable older adults. Therefore, the results may not be generalizable to all older adults with hypertension, particularly those who are frail or reside in nursing homes or long-term care facilities. PPH, defined using changes in SBP from before a meal to either 1 or 2 hours after the end of the meal, was not associated with an increased risk of a fall. Although the outcome in the current analysis was falls occurring at any time and not only after meals, it is possible that these postprandial windows do not fully capture the physiological processes that predispose older adults to falls.

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PERSPECTIVES

In the current study, PPH determined by ABPM was common among community-dwelling older adults with hypertension taking antihypertensive medication, but it was not associated with fall risk. The findings of the current study do not support that PPH or postprandial BP decline assessed by ABPM are effective in identifying older adults at elevated risk for falls.

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K. Reynolds and D. Shimbo had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Concept and design: K. Reynolds, D. Shimbo, C.B. Bowling, J.E. Schwartz, and P. Muntner. Acquisition, analysis, or interpretation of data: All authors. Drafting of the article: K. Narita and D. Shimbo. Critical review of the article for important intellectual content: All authors. Statistical analysis: R. Wei, L. Qian, and Zhou. Obtained funding: K. Reynolds and D. Shimbo. Administrative, technical, or material support: K. Reynolds, D. Shimbo, K. Cannavale, and T.N. Harrison. Supervision: K. Reynolds, D. Shimbo, K. Cannavale, and T.N. Harrison.

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Disclosures

Dr Narita has a grant from International Medical Research Foundation. D. Shimbo is a consultant for Abbott Vascular, Edward Lifesciences, Medtronic, and Tryton Medical. He conducts event ascertainment for trials that they are conducting (ie, testing coronary and valvular disease interventions). These studies are not related to the content area of the current article. The other authors report no conflicts.

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